

PUSHKAR SINGH

Bangalore, India · Open to Visa Sponsorship

☎ +91-9410560129

✉ singh.pushkar7570@gmail.com

🌐 [linkedin.com/in/pushkar-singh1/](https://www.linkedin.com/in/pushkar-singh1/)

🐙 github.com/pushkar7570/

Summary

AI/ML Engineer with 2+ years of experience building scalable machine learning and data-driven systems for enterprise workflows. Skilled in Python, SQL, data pipelines, and deploying AI models using RAG and LangGraph. Experienced in large-scale data processing, model evaluation, and developing robust, production-grade systems with MLOps practices.

Experience

PwC - PricewaterhouseCoopers

Feb. 2024 – Present

Associate 2 (Gen AI)

• ACTA - Agentic Controls Testing Assistant

- Designed and implemented ACTA as a multi-agent LangGraph-based system, orchestrating parallel agents using RAG and semantic indexing to process large-scale structured and unstructured datasets and automate audit workflows.
- Built and deployed on Azure using LangChain and data pipelines to extract, preprocess, and transform enterprise data, enabling efficient data flow, scalable processing, and reducing manual effort by 60%.
- Implemented model evaluation and confidence scoring pipelines to monitor performance, analyse error patterns, and optimize outputs, improving accuracy by 40% while reducing latency through retrieval tuning.

• CLUTS - Custom GPT based solution

- Built CLUTS, a GPT-powered pipeline with multi-step reasoning using prompt engineering and RAG to process and normalize large datasets, perform data preprocessing, and generate structured outputs across 5+ client engagements.
- Deployed on Azure using LangChain and API-based pipelines to analyse control data, generate test strategies, and optimize workflows, enabling scalable data-driven decision-making and improved operational efficiency.

Insignia Consultancy Solutions

May 2023 – Aug. 2023

Microservices Developer Intern

- Developed Spring Boot microservices for subscription management, implementing validation, transaction handling, and scalable backend services for reliable application performance.
- Developed REST endpoints, creating JPA entities and optimizing MySQL queries for accuracy and performance.

Education

Graphic Era Hill University, Dehradun

Sep. 2020 – Jun. 2024

B.Tech. in Computer Science & Engineering (8.25/10)

Maria Assumpta Sr. Sec. Convent School, Kashipur

Mar. 2019 – Apr. 2020

Senior Secondary Education - XII (85%)

Skills

- Programming & Systems:** Python, Java, SQL, Data Structures, REST APIs, Microservices, Distributed Systems
- AI & Machine Learning:** Scikit-learn, Model Training/Evaluation, RAG, LLMs, LangChain, LangGraph, CrewAI
- Data Engineering:** Data Pipelines, ETL, Pandas, NumPy, Large-scale Data Processing, FAISS, PineCone
- MLOps & Cloud:** Docker, CI/CD, Model Monitoring, Azure, AWS, API Deployment, FastAPI

Projects

Credit Risk Prediction for Loan Approval (Python, Scikit-learn, Pandas, NumPy)

- Built a predictive model to assess loan default risk using financial data, performing preprocessing, feature engineering, and exploratory analysis on structured datasets.
- Trained classification models achieving strong ROC-AUC and precision-recall metrics for scalable risk prediction.

Roxy - Interview Voice Assistant (Llama, FastAPI, RAG, LangGraph, ChromaDB, Whisper, Coqui)

- Built and deployed using FastAPI, ChromaDB, Whisper, and Coqui TTS, leveraging RAG, embeddings, and data preprocessing techniques to process multimodal data & generate retrieval-backed responses.
- Built and deployed on cloud using FastAPI, ChromaDB, Whisper STT and Coqui TTS, leveraging RAG, embeddings to process PDFs, audio, and video, enabling voice-driven interviews with real-time evaluation and retrieval-backed responses.
- Implemented self-improving pipeline that refines prompts, retrieval, and evaluation using feedback-driven workflow analysis.

Certifications & Publications

- Selected for **McKinsey Forward Program**, focused on problem solving, digital, and leadership skills.
- AWS Certified Cloud Practitioner**
- Published research** on breakout/reversal trading using candlestick-close and RSI divergence strategies. **IEEE Xplore**
- Awarded **PwC's Extra Miler Award & Quarterly LoS** for taking Agentic AI initiatives and project deliverables.